

Education

- (expected) 05/26  **Ph.D., University of Michigan: Statistics**
5th year Ph.D. candidate working under Ambuj Tewari on the theory of deep learning.
- 05/25  **M.A., University of Michigan: Statistics**
Received Master's degree while working towards my Ph.D.
- 05/20  **B.S., University of Illinois Urbana-Champaign: Computer Science and Statistics, Mathematics**
Graduated Cum Laude with Highest Distinction in both majors. GPA: 3.88.

Research Publications

- 1 **J. Trauger** and A. Tewari, "On next-token prediction in llms: How end goals determine the consistency of decoding algorithms," *arXiv preprint arXiv:2505.11183*, 2025,
A poster of this work won best poster at the Midwest Machine Learning Symposium 2025.
- 2 **J. Trauger**, T. Trauger, and A. Tewari, "Characterizing the multiclass learnability of forgiving 0-1 loss functions," *arXiv preprint arXiv:2510.08382*, 2025.
- 3 **J. Trauger** and A. Tewari, "Sequence length independent norm-based generalization bounds for transformers," in *International Conference on Artificial Intelligence and Statistics*, PMLR, 2024, pp. 1405–1413.
- 4 K. M. Ikegwu, **J. Trauger**, J. McMullin, and R. J. Brunner, "Pyif: A fast and light weight implementation to estimate bivariate transfer entropy for big data," in *2020 SoutheastCon*, IEEE, 2020, pp. 1–6,
Package can be found [on PyPI](#).

Employment History

- 8/21 – present  **Graduate Student Instructor: University of Michigan**
Helped create curriculum, teach labs, hold office hours, and grade for various classes in the Statistics department at the University of Michigan. *Won honorable mention for best GSI team in the Statistics department in the 2022-2023 school year.* Classes taught include: Introduction to Statistics, Honors Introduction to Statistics, Applied Statistical Methods II, Applied Regression Analysis, and Statistics & Artificial Intelligence.
- 8/20 – 8/21  **Data Science Engineer: SimSpace**
Part of a team that focused on creating a reinforcement-learned agent to emulate a human hacker. I helped design and implement code for the agent, the simulator, and how other teams would interact with the agent. I was also tasked with ML scientist projects (e.g., designing the rewards to make the agent act like a human hacker at various levels of expertise.) Helped in the interview process by designing questions and evaluating candidates for software engineering positions. Was responsible for managing the in-house networks we would use to test our agents.
- 5/20–8/20  **Data Science and Machine Learning Infrastructure Researcher: Gies College of Business' Data Science Research Service**
Used Microsoft Azure to support employees with their computational needs. Also was a data science/ML consultant for students, faculty, and staff.